

Trader Behaviour under Bitcoin Fear & Greed Sentiment

1. Objective

- Understand how **trader performance** (PnL, win rate) changes with **market sentiment** (Fear & Greed Index).
- See how **trading behaviour** (number of trades, position size, fees) differs between **Extreme Fear, Fear, Neutral, Greed, Extreme Greed**.
- Compare **top-performing** and **bottom-performing** traders under these sentiment regimes.
- Look at **trends over time**: how sentiment, activity, and PnL move together.

2. Data & Features

- Historical trade data aggregated to **account-day** level.
- Engineered metrics per account-day:
 - total_daily_pnl: net PnL for the day.
 - win_rate: share of winning trades.
 - num_trades: number of trades.
 - avg_size_usd: average trade size in USD.
 - avg_fee_usd: average fee per trade.
- Joined with **daily Fear & Greed Index (value)** and a **sentiment_class** label: Extreme Fear, Fear, Neutral, Greed, and Extreme Greed.

	Account	trade_date	total_daily_pnl	num_trades	wins	avg_size_used	avg_fee	win_rate	value	sentiment_class
0	0x083384f897ee0f19899168e3b1bec365f52a9012	2024-11-11	0.000000	177	0	5089.718249	0.948000	0.000000	76.0	Extreme Greed
1	0x083384f897ee0f19899168e3b1bec365f52a9012	2024-11-17	0.000000	68	0	7976.664412	0.998288	0.000000	90.0	Extreme Greed
2	0x083384f897ee0f19899168e3b1bec365f52a9012	2024-11-18	0.000000	40	0	23734.500000	2.373450	0.000000	83.0	Extreme Greed
3	0x083384f897ee0f19899168e3b1bec365f52a9012	2024-11-22	-21227.000000	12	0	28186.666667	2.818666	0.000000	94.0	Extreme Greed
4	0x083384f897ee0f19899168e3b1bec365f52a9012	2024-11-26	1603.100000	27	12	17248.148148	3.874907	0.444444	79.0	Extreme Greed
...	...	...	...	...	...	...	...	...	...	...
2336	0xbec1707d6b44d4d52bfe19e41f8a828645437aab	2025-04-27	1709.194807	82	32	2153.859390	0.312681	0.390244	61.0	Greed
2337	0xbec1707d6b44d4d52bfe19e41f8a828645437aab	2025-04-28	4008.588908	430	169	1495.538419	0.371047	0.393023	54.0	Neutral
2338	0xbec1707d6b44d4d52bfe19e41f8a828645437aab	2025-04-29	8561.771838	902	398	1939.739989	0.386831	0.441242	60.0	Greed
2339	0xbec1707d6b44d4d52bfe19e41f8a828645437aab	2025-04-30	2520.773814	75	39	1748.805333	0.270870	0.520000	56.0	Greed
2340	0xbec1707d6b44d4d52bfe19e41f8a828645437aab	2025-05-01	1364.022527	131	43	2448.466489	0.457253	0.328244	53.0	Neutral

2341 rows x 10 columns

3. Methods (very brief)

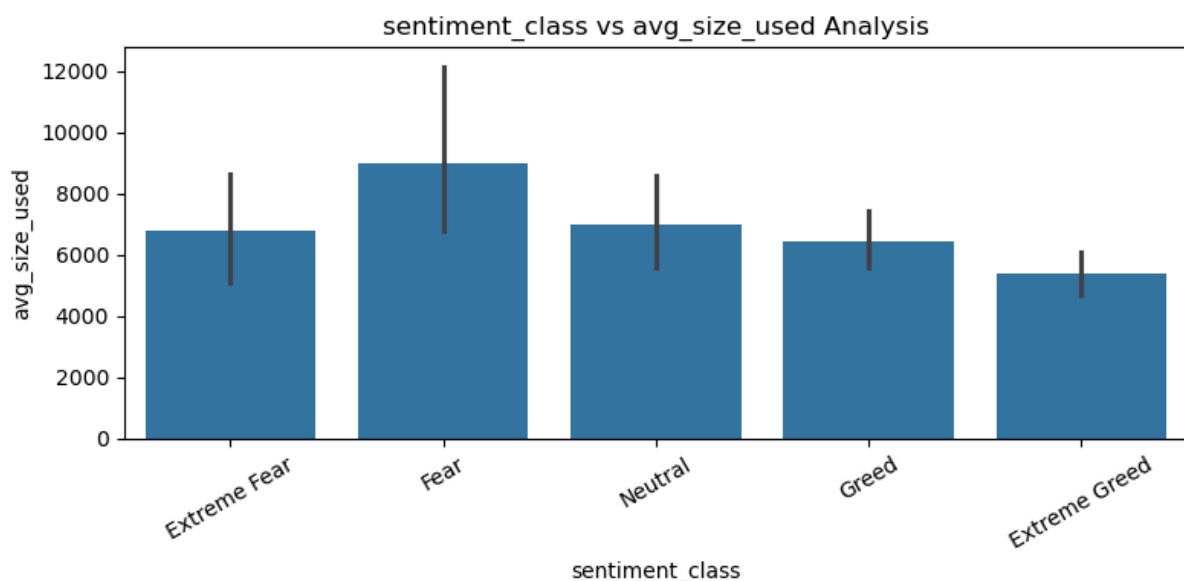
- Used **Python (pandas, Seaborn)** in notebook_2.
- Grouped data by **sentiment_class** and by **Account + sentiment_class**.
- Created **Top** and **Bottom** trader groups based on **average daily PnL**.
- Calculated **correlation** between numeric features.
- Built **daily time-series** and used a **7-day rolling average** to smooth PnL, sentiment, and number of trades.

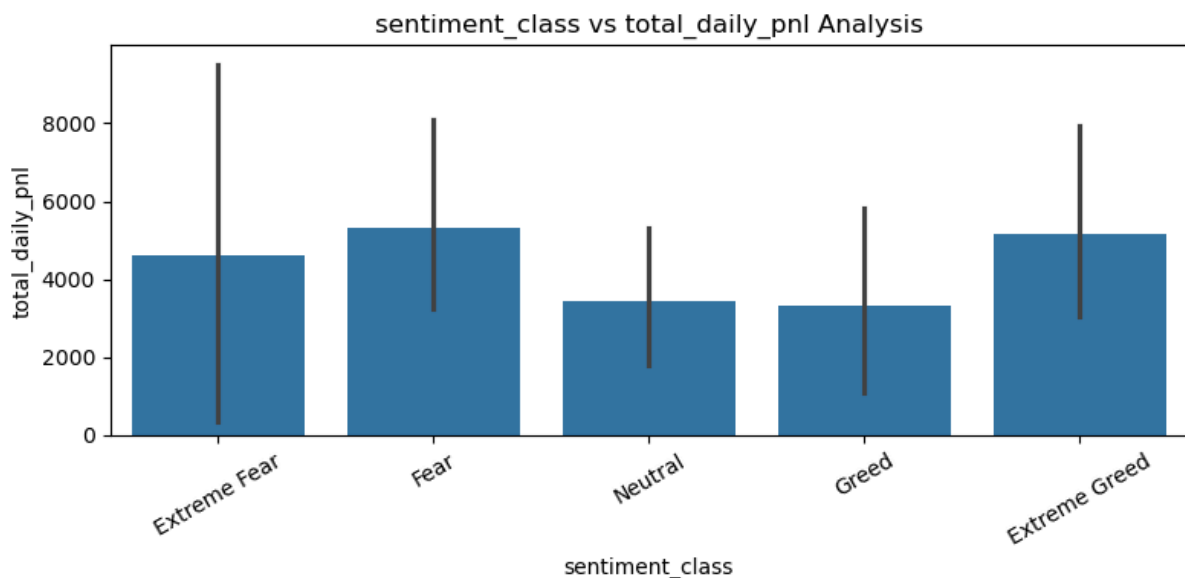
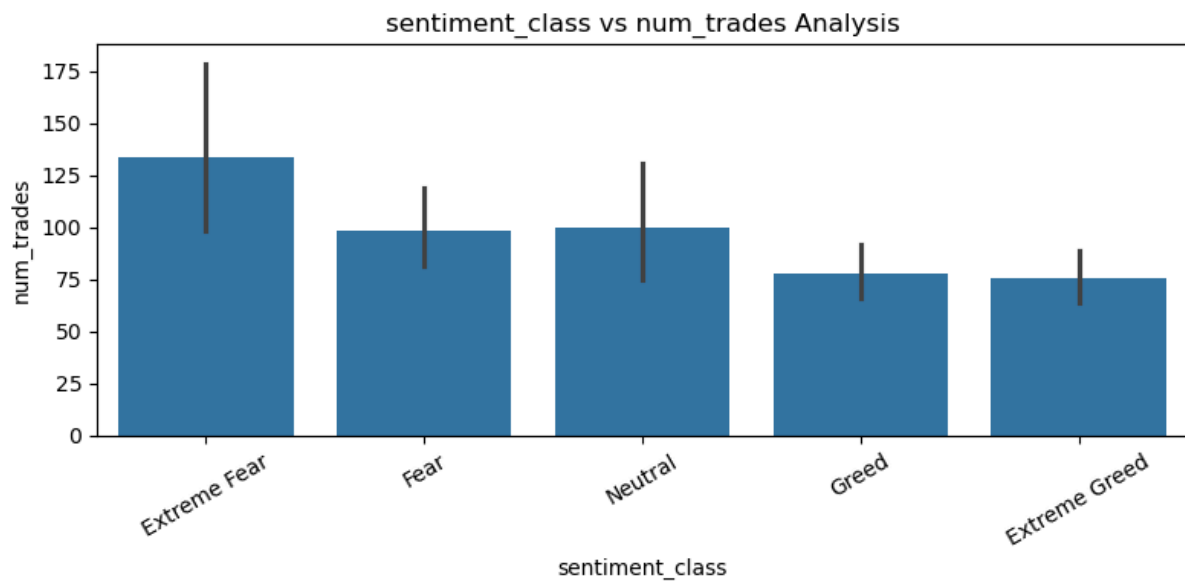
5. Main Findings

5.1 Behaviour by sentiment class

Bullets:

- On **Extreme Fear** days, traders do **more trades per account**, and PnL and win rate are **very volatile**.
- On **Extreme Greed** days, traders take **fewer trades** with **smaller average size and fee**, but show **higher win rate and solid PnL**.
- **Neutral** and **Greed** days are in the **middle** for most metrics.
- **Per-account performance and risk-taking**



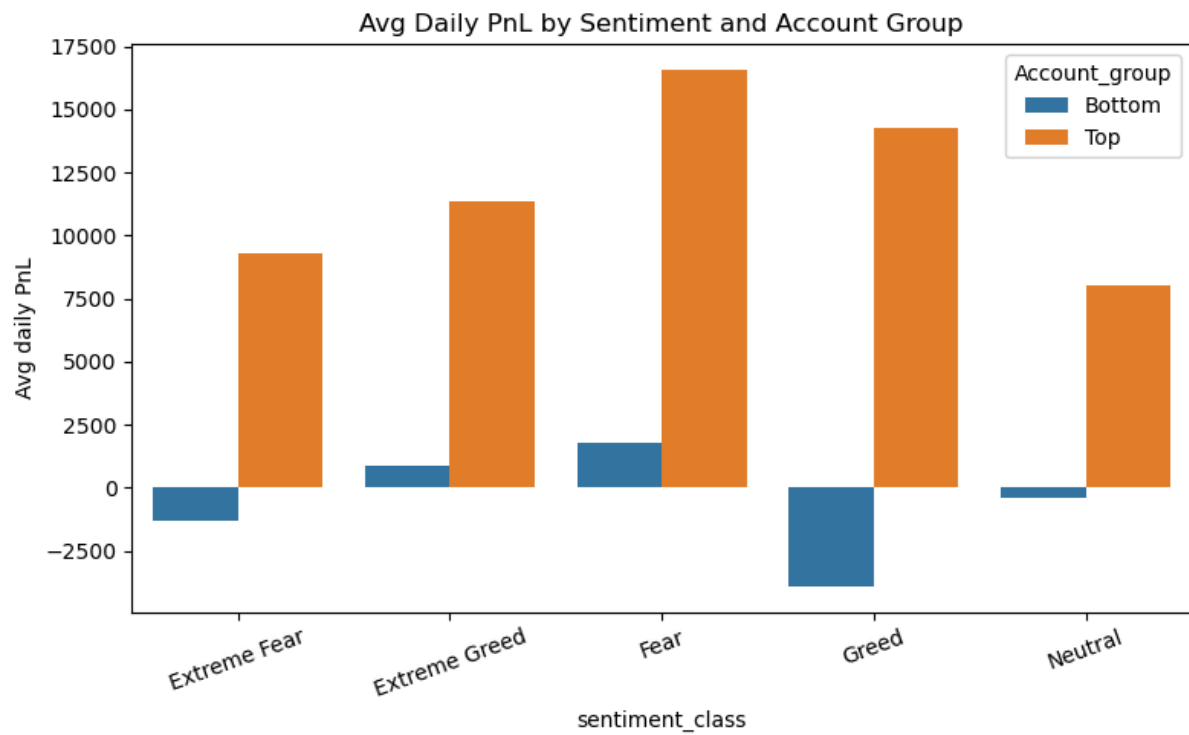


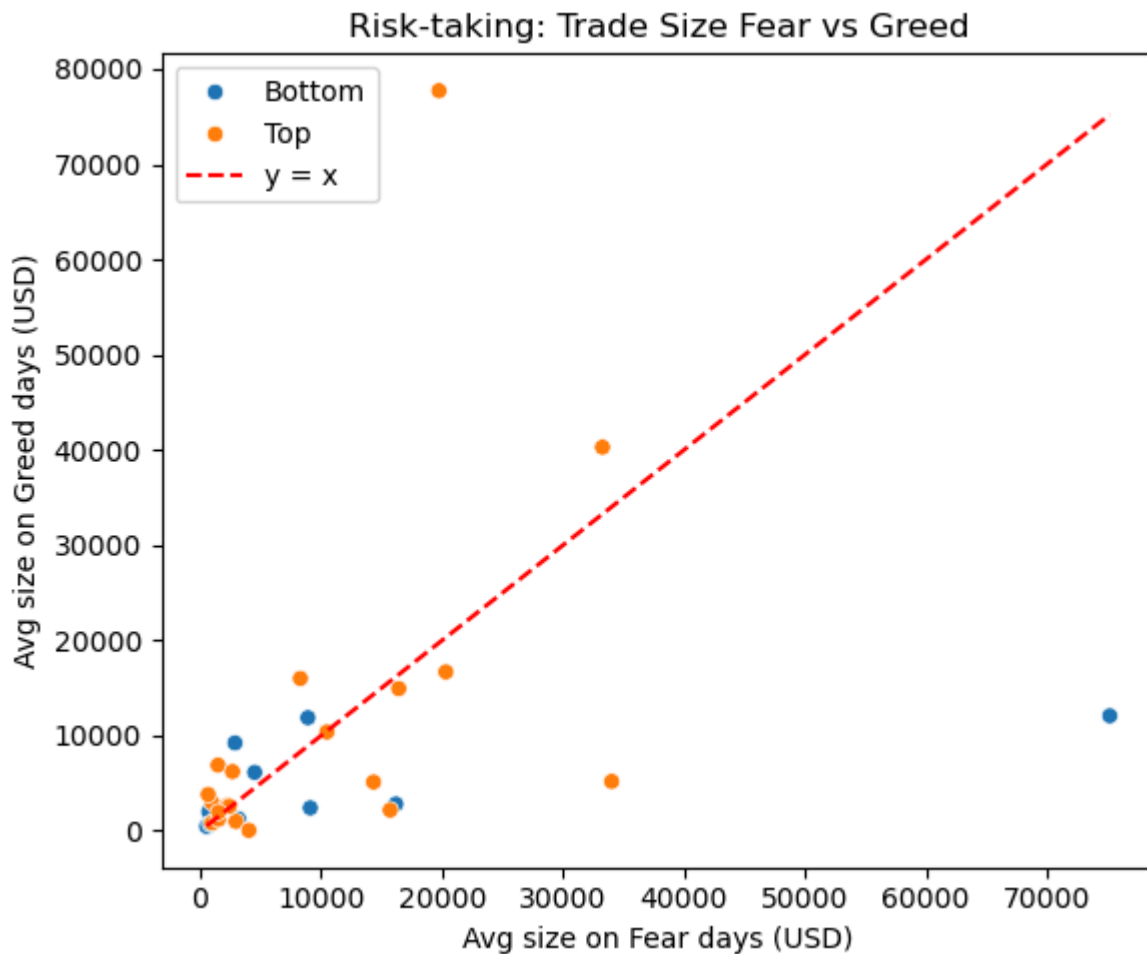
5.2 Top vs Bottom traders

Bullets:

- Accounts were ranked by overall average daily PnL into **Top** and **Bottom** groups.
- **Top traders** are profitable in **all** sentiment classes, including Fear and Extreme Fear.
- **Bottom traders** lose money in **every** sentiment class, even during Greed and Extreme Greed.

- Many Top traders **increase position size and sometimes trade count on Greed days**, while Bottom traders keep **small and relatively stable activity**.

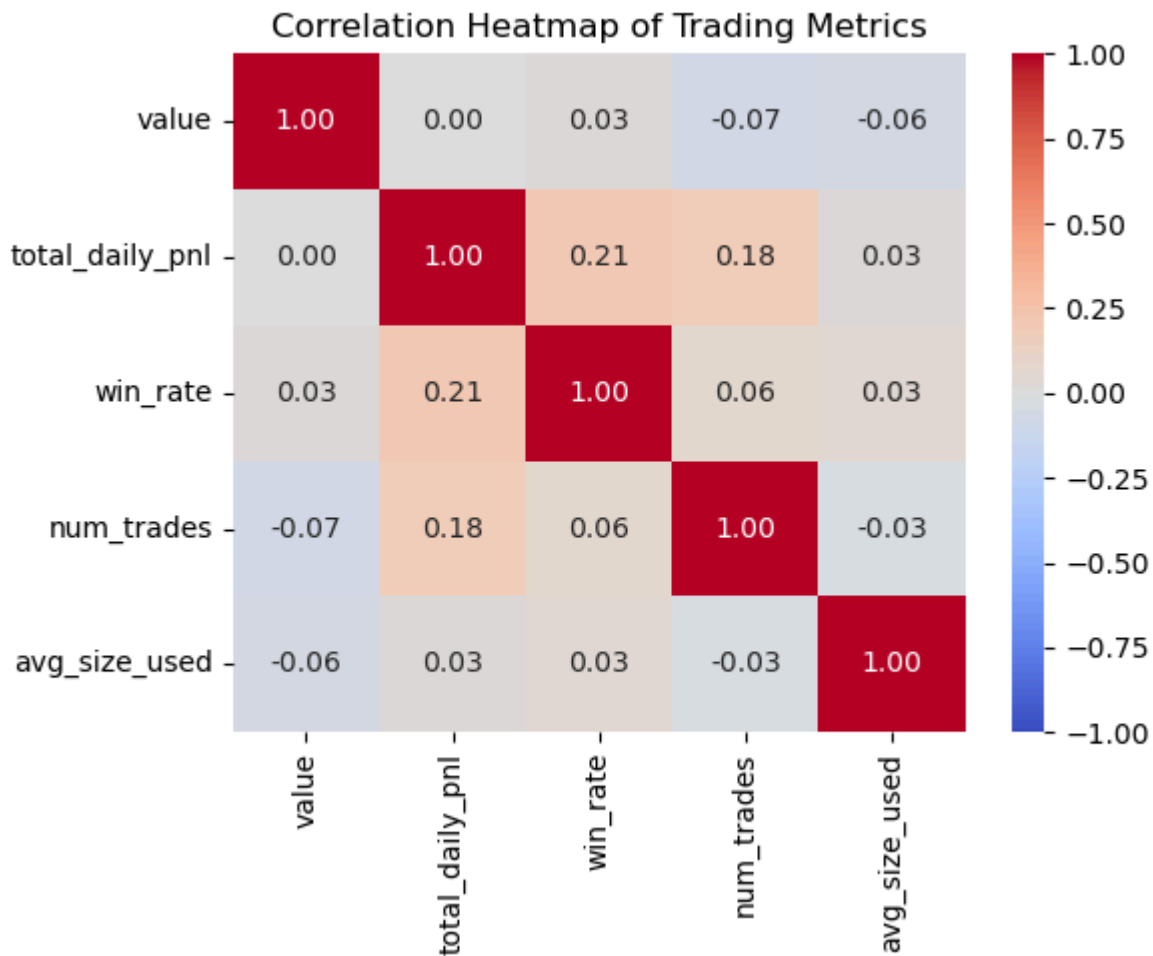




5.3 Correlation

Bullets:

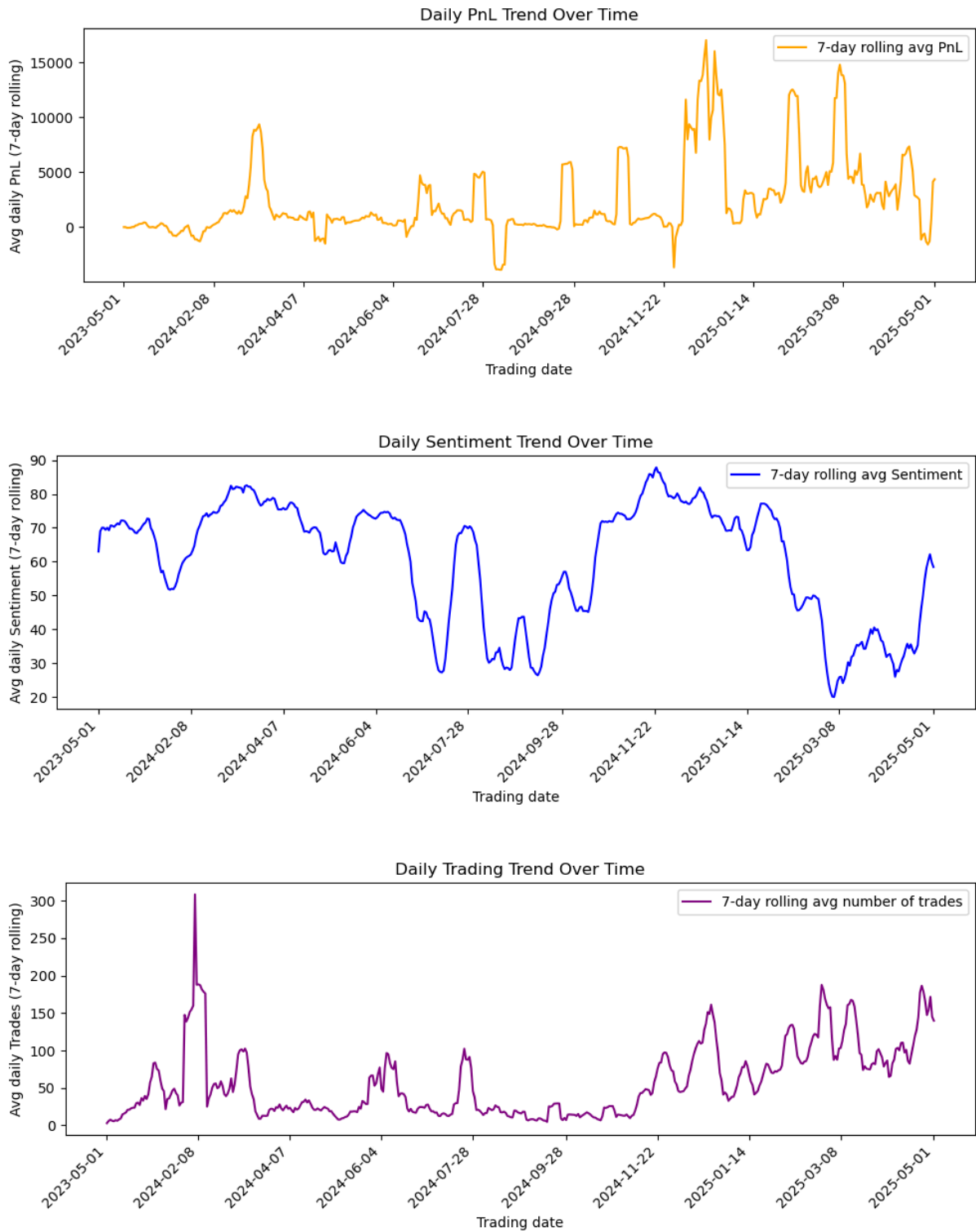
- The correlation matrix shows **no strong relationships** between sentiment value, PnL, win rate, number of trades, and average size (all coefficients are small).
- Example: sentiment value vs PnL and sentiment value vs number of trades both show **almost flat scatter plots**, confirming weak linear relationships.



5.4 Time trends

Bullets:

- The **7-day rolling average PnL** is small and choppy through much of 2023–mid-2024, with short profit spikes.
- From late 2024 to early 2025, rolling PnL becomes **much higher**, with several strong peaks.
- During this period, the **rolling sentiment value** is mostly in **Greed/Extreme Greed**, and the **rolling number of trades** is also higher.
- When rolling sentiment drops into strong Fear, both PnL and trading activity tend to fall or become more unstable.



6. Limitations & Future Work

Bullets:

- This is **exploratory analysis only**; it does not build prediction models.

- Data covers a limited time period and market; results may not generalize to all assets.
- Possible next steps:
 - Use regression or ML models to **quantify** impact of sentiment and behaviour features on PnL.
 - Cluster accounts to find more detailed **trader types**.
 - Back-test strategies that change position size or trade frequency based on sentiment regime.